

```
/usr/local/trouble-maker/bin/trouble-maker.pl
--version=RHEL_5 --selection=do_nothing.tar
```

Now run the command shown below:

```
/usr/local/trouble-maker/bin/trouble-maker.pl
--version=RHEL_5
```

...and wait for things to explode. Once you run it, you'll get a description about what has gone wrong. Your job is to deal with the problem and fix it. Run the command and restart the system to see what happens. You'll come across various cases on RHEL5 while playing with this tool. Twelve such scenarios are discussed here.

Scenario 1: 'You have logged into the command line mode instead of GUI mode'

Here you see that once you've restarted your machine, it starts in command line mode instead of the default GUI mode. This is a clear indication of a problem in the *inittab* file. Your default runlevel is specified in the */etc/inittab* file. If you open this file, you'll see a line *id:3:initdefault:* which is the root cause of this problem. This tells Linux to boot into runlevel 3 (command line mode). Change it to runlevel 5 (for GUI mode), save the file and restart your machine. Next time you'll be back on your GUI mode. Before you run Trouble-Maker again, don't forget to delete the */tmp/trouble-maker/* backup and */tmp/trouble-maker/rescue* directories.

Scenario 2: 'INIT: no more processes left in this runlevel'

This is again a problem with your *inittab* file. This file contains information about the processes to be run at each runlevel. You can find the solution for this problem in one of my previous articles at <http://www.linuxforu.com/2013/04/learn-the-art-of-linux-troubleshooting/>. Here, just look at the section that describes what to do when your *inittab* files are deleted. Though our *inittab* file has not been deleted but, rather, is misconfigured, we'll still need to reinstall the *initscripts* package.

Scenario 3: 'The GDM user 'gdm' doesn't exist'

User information is stored in the */etc/passwd* file and this message shows that there is something wrong with that file. When you click on OK and you try to log in at the command line, you will still find problems. This is the time to check your */etc/passwd* file and you can find the solution in the article mentioned earlier (link given). In fact, I suggest that you read that article along with <http://www.linuxforu.com/2013/03/playing-hide-and-seek-with-passwords/> for some valuable solutions to such situations.

Scenario 4: 'Can't log in as the root'

When you run Trouble-Maker and get this message as the problem description, restart your machine. If you can

normally log in to your machine, it means your *passwd* file is intact. It's time to switch to another terminal and check whether you can log in there or not. If, when you enter your username, instead of asking for your password, it repeatedly shows the localhost login message, look for the */etc/pam.d/login* file. If it's missing, then refer to the two articles I've mentioned earlier to get the solution.

Scenario 5: 'switchroot:mount failed: No such file or directory'

If you get this message after rebooting your machine, it shows that the location of the root that you have specified is invalid. To solve this problem, you'll have to go to rescue mode and open */boot/grub/grub.conf* file. Here, edit the line about the kernel and point to your valid root location. Again, the articles mentioned earlier will be handy as they give a step-by-step explanation about troubleshooting in single user mode.

Scenario 6: 'initrd file not found'

Earlier it was the root location and now it's the *initrd* file name. The process of rectifying this is quite similar to what was done in Scenario 5. You'll have to enter rescue mode and then use the following command:

```
ls /boot .
```

Here you'll get your *initrd* file name. Now open *grub.conf* file and correct the corresponding entries. The next time you boot, everything will be fine.

Scenario 7: 'Can't log in to the root via a console'

When you get this message as the Trouble-Maker problem description, whenever you try to log in via another terminal, it won't let you do so. So check your */etc/pam.d/login* file as we did earlier and, if it's intact, then there's a good chance that your */etc/securetty* file has some issues. If it's so, then make the necessary amendments; save the file, restart the machine and check whether the problem has been solved or not.

Scenario 8: 'Network is not working'

If you get this problem description after running Trouble-Maker, the stage is all set for some network related issue. After restarting, fire the *ifconfig* command to check whether you've got an IP address. Here, you won't see any IP address. Then check the status of the network service by issuing the *service network status* command. If you don't get any output, then the networking facility is not available within your system. This is controlled by */etc/sysconfig/network* file. Open it and if you see *NETWORKING=no* then set it to yes. Save the file and restart the network service. Now, if you issue *ifconfig*, you'll get your IP address.